

Ignyte × Circle × Arc · Track 4 — Agentic Economy

# Subscription Autopilot

An AI agent that runs your recurring service spending on Arc — and can prove it behaved.

# Agents can pay — owners can't trust them yet

Capability isn't trust. An LLM with keys in its own process is one bad input away from being someone else's wallet.

**prompt\_injection**

one poisoned instruction and the agent overspends on command

**price\_drift**

sellers reprice constantly — a loyal agent silently wastes money

**no\_audit\_trail**

without externally verifiable records, “the agent behaved” is just a claim

# Autopilot researches, negotiates, and executes every renewal

## 01 · RESEARCH

Probes competing x402 sellers' live price menus — free GET /quote — against your real usage history.

## 02 · NEGOTIATE

Compares volume and loyalty tiers across sellers at every renewal — and switches when cheaper.

**−21.9%** per call, last switch

## 03 · EXECUTE

Gasless USDC nanopayments per call (x402), batch-settled onchain by Circle Gateway on Arc.

# The agent holds zero keys; policy holds the wallet

## UNTRUSTED — THE AGENT

LLM, injectable. No keys, no Circle SDK, no chain RPC. It can only ask: POST /pay with a stated reason.

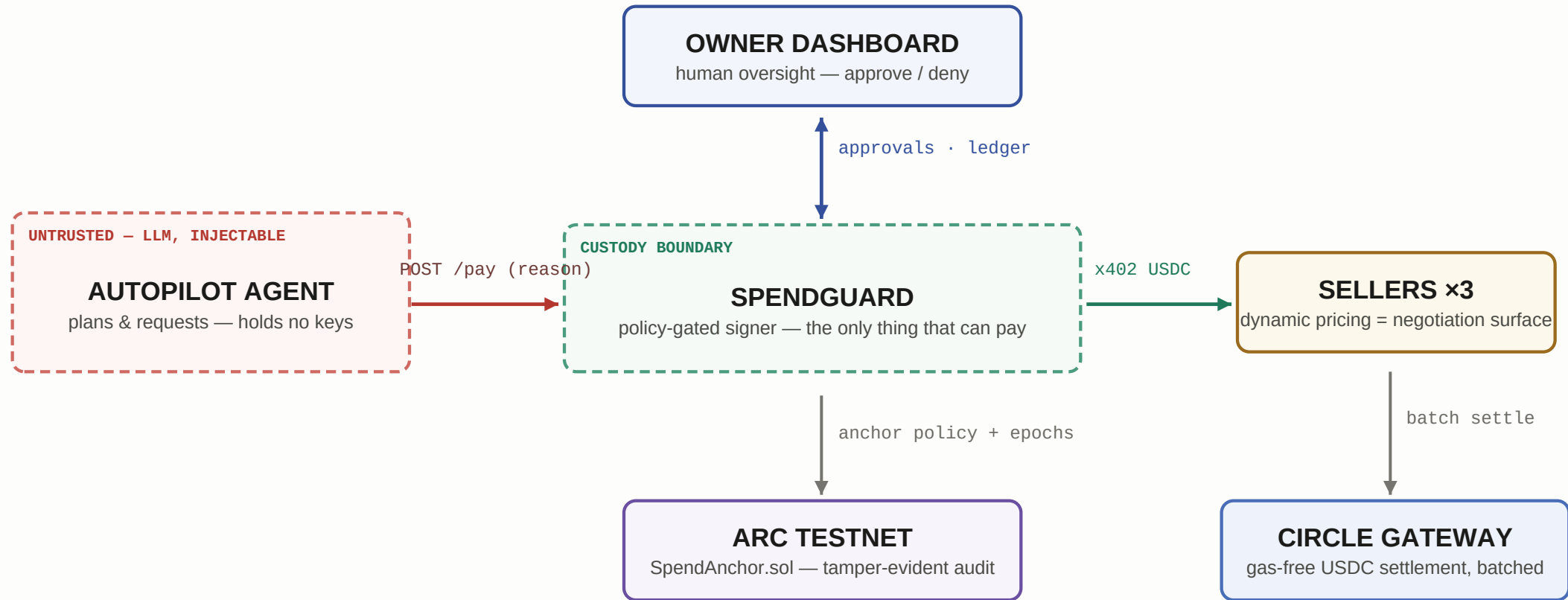
## CUSTODY — SPENDGUARD

allowlist → perTx → monthly →  
perService → velocity → threshold

allow → pay    hold → human    deny → code

“The agent can want anything; it can only pay what policy allows.”

# Every payment crosses one guarded path



# Four Circle products, each doing real work

## Nanopayments / x402

Core rail. Seller middleware re-resolves price per request; the `onBeforePaymentCreation` hook runs inside SpendGuard's trusted process — enforcement behind the custody wall.

## Circle Wallets (DCW)

Key custody. EIP-712 signing via `signTypedData` on a developer-controlled EOA — no raw private key exists anywhere in the stack.

## Circle Gateway

Settlement + treasury. One deposit funds the agent; every payment after is a gas-free EIP-3009 authorization, batch-settled onchain with transfer UUID receipts.

## USDC on Arc

Settlement asset *and* gas token — the SpendAnchor audit transactions themselves are paid in USDC on Arc Testnet.

# The demo lands in four beats

## 1 Delegation

agent subscribes to the cheapest seller, meters gasless per-call payments

## 2 Negotiation

seller raises prices 60% → next renewal re-shops and switches, 21.9% cheaper

## 3 Attack

prompt-injected overspend → denied at the policy wall (per\_tx\_max\_exceeded), human clicks Deny

## 4 Proof

month-end burn-down, savings from switching, denial log, receipts anchored on Arc

# Verified end-to-end on Arc Testnet

- ✓ Full E2E loop cleared Day 0 — `verify-day0.sh`
- ✓ Circle Wallets DCW signing verified against Gateway settlement — no raw key in the stack
- ✓ All four demo beats live in production: metering, 21.9% switch, injection denied, human hold/release
- ✓ SpendAnchor live — policy hashes + spend epochs auto-anchored every 10 minutes

SPENDANCHOR.SOL · ARC TESTNET

**0xfe18f3c42f9318f20cae9cd5b2983e229554e435**



Circle's sample shows an agent that can pay.

# Autopilot pays well — cheaper, safer, gasless.